

MANE 4140 – Intro to CFD

Midterm Project Report

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Effort Percentage: 30% Mergen, 40% Povie, 30% Sonawane

Section 1: Problem Description

The goal of this experiment is to analyze the aerodynamic performance of a NACA 0015 airfoil and a NACA 0015 airfoil with a slat by measuring the coefficient of lift (C_L), and coefficient of drag (C_D) at freestream velocities (45 m/s and 90 m/s) and angles of attack (0° & 12°). The study aims to evaluate the effects of slat implementation on aerodynamic efficiency under steady, turbulent, and incompressible flow conditions in a 2D simulation with a chord length of 0.254m (10in). The chord length of the slat is $0.3c = 0.072$ m. The slat airfoil is also a NACA0015.

The flow is measured at velocities of 45 m/s ($Re = 7.83e5$) and 90 m/s ($Re = 1.57e6$), with the flow being steady, turbulent, and incompressible at angles of attack (AOA) 0 degrees and 12 degrees. The material input is air at sea level at 15 deg Celsius with a density of 1.225 kg/m^3 and a viscosity of $1.781e-5 \text{ Pa}\cdot\text{s}$.

Section 2: Problem Setup

The boundary conditions include a no-slip condition applied around the airfoil geometry with a wall velocity of 0, where a wall-resolved approach is used for near-wall treatment with a roughness height of 0. A slip condition is applied to the $\pm z$ surfaces for the 2D simulation. Along with these, a far-field boundary condition is applied to the circular rim with freestream velocities of 45 m/s and 90 m/s. The physics of this study involves a steady-state simulation with incompressible turbulent flow assumptions. The turbulence model used is Spalart-allmaras.

Section 3: Numerical Scheme and Meshing

Numerical Scheme:

All default settings from Acusolve were used for all meshes and solutions

Mesh Setup:

For 0015 airfoil with no slat (basic) and both 0 deg and 12 deg cases the following mesh setup was used:

- Edge mesh with average element size set to $c/128 = 1.984375e-3$ m
- Edge layer with parent surface as the +z face. First layer thickness set to $c/300,000 = 8.4667e-7$, with 43 layers. Constant growth method at a rate of 1.2 and termination policy set to truncate
- A box refinement zone was made with x,y,z measurements of $0.2c = 0.5588$ m, $0.3c = 0.0762$ m, and $0.3c = 0.0762$ m. The average element size was set to $c/128 = 1.984e-3$ m. For the 12 deg cases, the box was rotated 12 deg ccw around the z axis, and the y measurement was changed to $0.5c = 0.128$ m
- Two circular refinement zones were used at the leading edge and trailing edge. The leading edge refinement zone has a radius of $c/32 = 7.9375e-3$ m and an element size of $c/1024 = 2.48e-4$ m. The trailing edge refinement zone has a radius of $c/128 = 1.984e-3$ m and the same element size as the leading edge zone.
- A surface mesh was ran on +z face with maximum mesh element size set to $c = 0.254$ m
- Extrusion performed with +z face as the source. Element size along extrusion was set to $c/10 = 0.0254$ m, and element type was set to quads
- Volume mesh was ran with maximum element size as $c = 0.254$ m

For 0015 airfoil with slat and both 0 and 12 degree cases the following mesh set up was used:

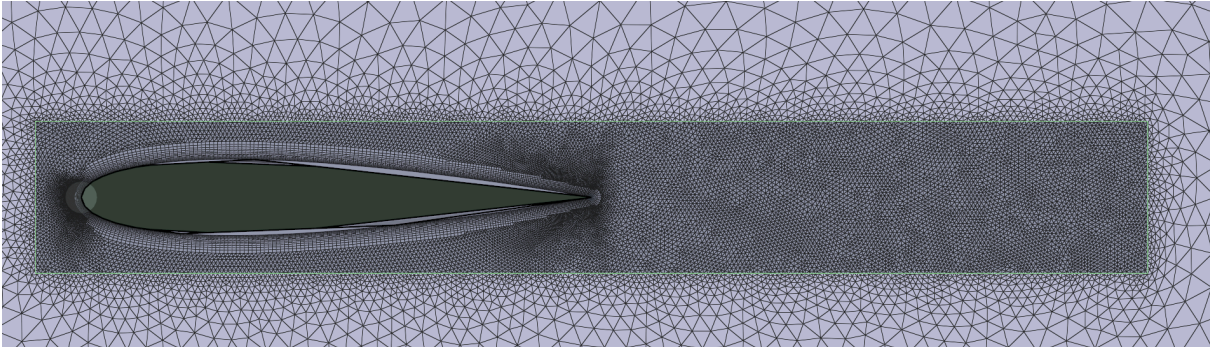
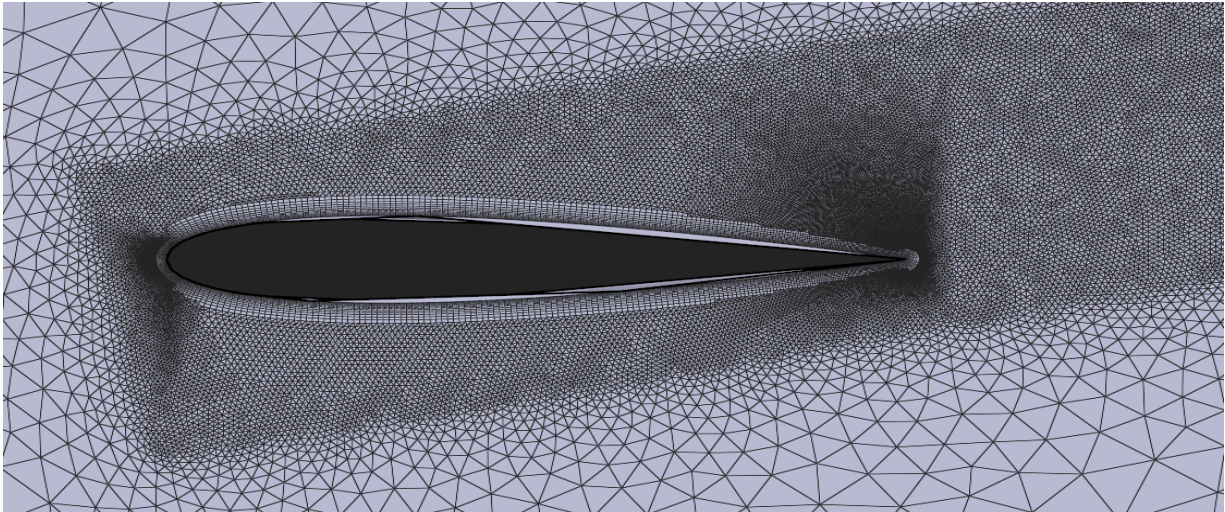
- Edge mesh with average element size set to $c/128 = 1.984375e-3$ m on the airfoil, and a similar edge mesh on the slat with element size set to $c/128 = 5.95e-4$ m
- Edge layer with parent surface as the +z face. First layer thickness set to $c/300,000 = 8.4667e-7$, with 43 layers. Constant growth method at a rate of 1.2 and termination policy set to truncate. Edge layer only applied to the airfoil.
- A box refinement zone was made with x,y,z measurements of 0.65 m, $0.5c = 0.128$ m, and $0.3c = 0.0762$ m. The average element size was set to $c/128 = 1.984e-3$ m. For the 12 deg cases, the box was rotated 12 deg ccw around the z axis. The box refinement zone covered both the airfoil and the slat
- Two circular refinement zones were used at the leading edge and trailing edge. The leading edge refinement zone has a radius of $c/32 = 7.9375e-3$ m and an element size of $c/1024 = 2.48e-4$ m. The trailing edge refinement zone has a radius of $c/128 = 1.984e-3$ m and the same element size as the leading edge zone.
- A surface mesh was ran on +z face with maximum mesh element size set to $c = 0.254$ m
- Extrusion performed with +z face as the source. Element size along extrusion was set to $c/10 = 0.0254$ m, and element type was set to quads
- Volume mesh was ran with maximum element size as $c = 0.254$ m

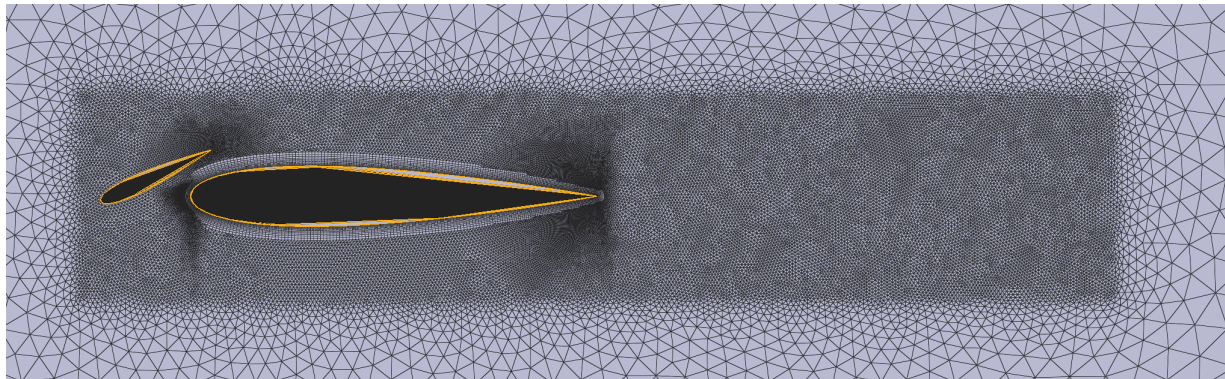
A summary of the volume mesh elements for each case can be seen below:

Volume Mesh Elements:	0 degree AOA	12 degree AOA
NACA 0015 without slat (basic)	182,196	249,946
NACA 0015 with slat	242,792	240,846

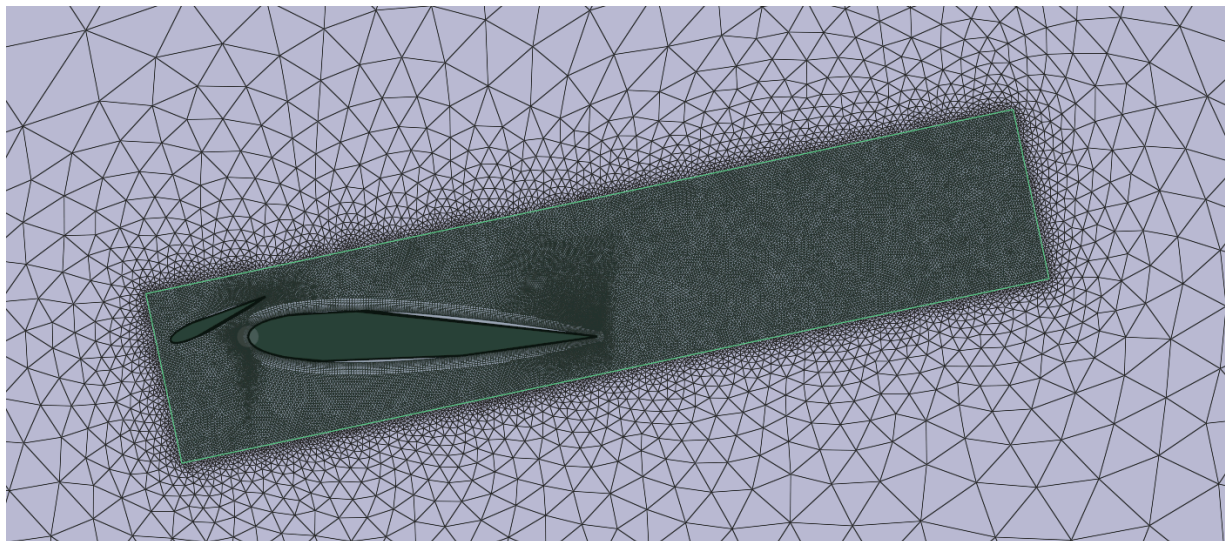
Screenshots of the meshes for each case can be seen in table 1:

Table 1: Meshes for each case

 0 deg without slat
 12 deg without slat



0 deg with slat



12 deg with slat

Section 4: Other details and issues

Additional details:

- For the 12 deg cases, solver controls were modified:
 - The steady state update factor was changed to 0.8
 - The steady maximum steps was increased to 200
- All other settings are default

Issues in project and meshing:

Adding the slat to the geometry posed many problems in meshing for the group. Due to the different chord lengths of the slat and the wing, adding refinement zones to the slat was very

difficult. In the location where the leading edge of the airfoil and the trailing edge of the slat are close together, the different element sizes in the refinement circles cause the meshes to overlap, which causes the software to have an issue when trying to run the solution. A fix to this problem was making the average mesh element size for the main airfoil's refinement zones smaller. Doing this created a mesh with over a million volume elements, and the team's computers couldn't handle this. The team could not find a solution for this problem, so to fix the issue, the team had to remove the refinement zone on the trailing edge of the slat.

The initial airfoil choice for the project was a NACA 4412 airfoil. After creating the geometry of the airfoil and running the initial mesh, an issue arose when trying to apply the extrusion to the surface mesh. The source of this error was unknown, and due to time constraints, the team decided to pivot to use a simpler, non-cambered airfoil.

Section 5: Results

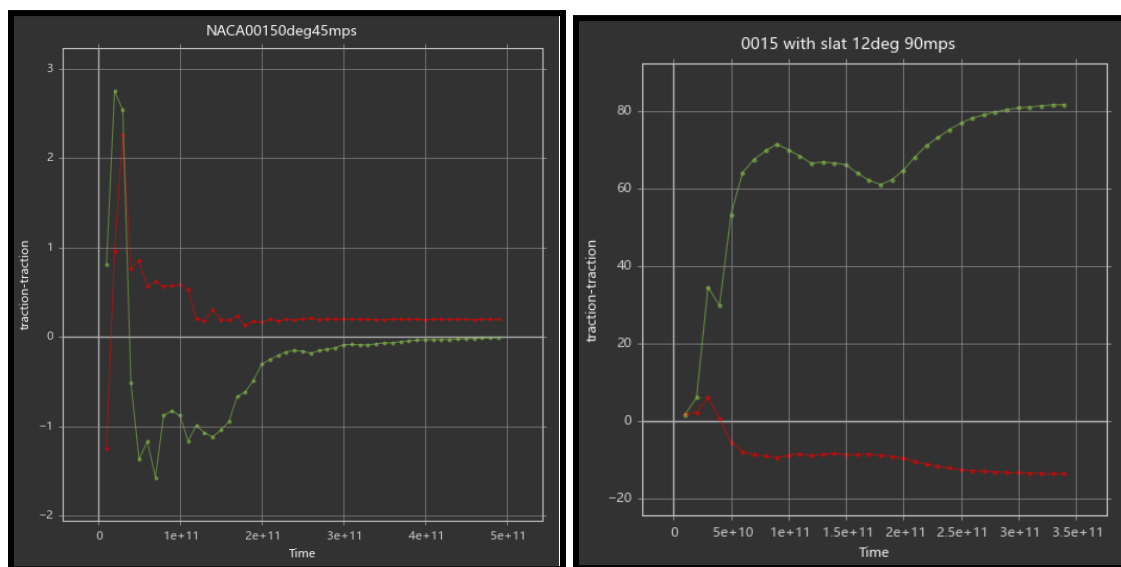


Figure 1: Traction forces on the NACA 0015 airfoil at 0 degree AOA at 45m/s and NACA 0015 airfoil with a slat at 12 degree AOA at 90m/s

From these plots the drag and lift forces for each case were found. The values were taken after a long time (i.e. stable slope of the plot), to capture the most realistic forces the wings would experience at each condition. The higher force peaks in the slat configuration compared to the airfoil at 0 degree AOA shows an increased lift due to the added slat geometry. This also means a higher drag force on the airfoil.

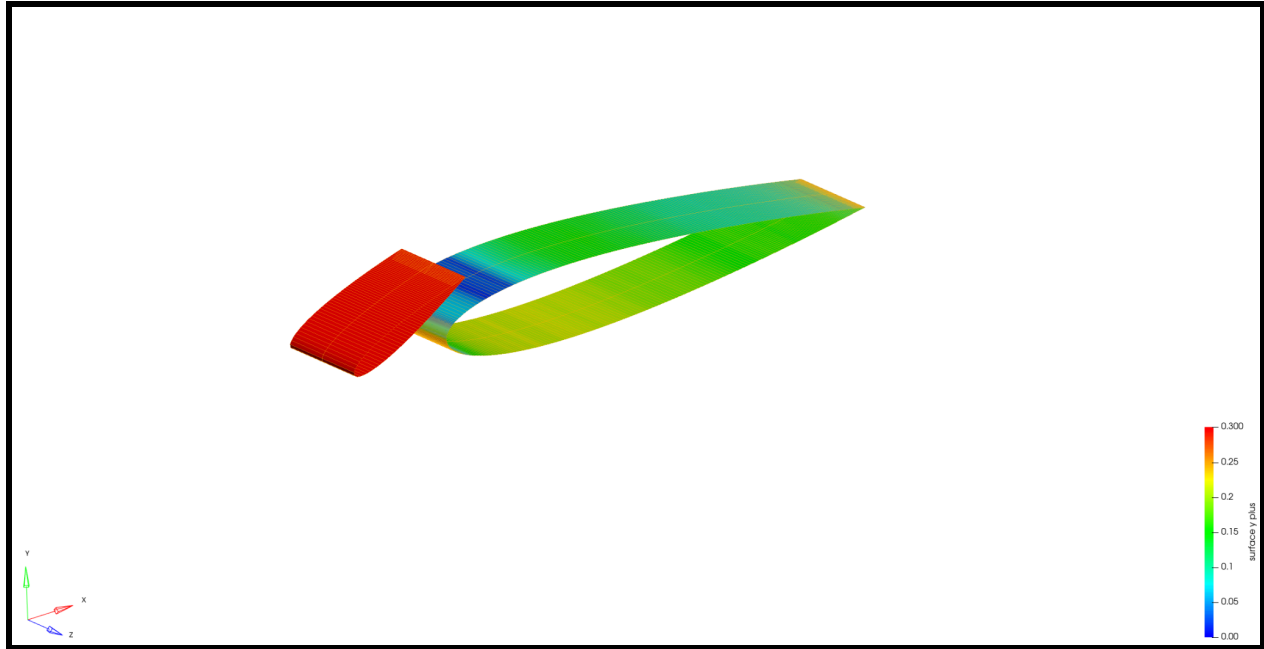


Figure 2: Surface y^+ of the NACA 0015 airfoil with a slat at 0 degree AOA at 90m/s (Every case had a main airfoil surface y^+ plus of less than 0.3)

Due to mesh issues, a surface y^+ plus did not have the resolution necessary to be accurate for the slat. Because the main airfoil is our point of interest, this shouldn't affect the results greatly.

Case	Drag (N)	Lift (N)
basic 0 deg 45m/s	0.1959	-0.0094
basic 0 deg 90m/s	2.0581	-0.1708
basic 12 deg 45m/s	3.150	18.462
basic 12 deg 90m/s	13.430	75.709
slat 0 deg 45m/s	1.0265	-1.640
slat 0deg 90 m/s	4.1049	-6.1383
slat 12 deg 45m/s	3.2298	19.959
slat 12 deg 90 m/s	13.467	81.642

Table 2: Lift and Drag Force values based of the 8 different cases

Lift and drag forces were pulled from traction force plots when the forces approached steady state. Through these and keeping standard air conditions in mind ($\rho = 1.23kg/m^3$), then dynamic pressure (q) is calculated as:

$$q = \frac{1}{2} \rho V^2$$

Where V is the freestream velocity (45m/s or 90m/s). Through dynamic pressure, the Lift coefficient (C_L) and Drag coefficient (C_D) given by

$$C_L = \frac{L}{qS}$$

$$C_D = \frac{D}{qS}$$

Where S is the area of the 2D airfoil given as the product of the chord * span length. The table with the C_L and C_D values is given in the table below

Case	Drag (N)	Lift (N)	C_D	C_L
basic 0 deg 45m/s	0.1959	-0.0094	0.0082	-0.00039
basic 0 deg 90m/s	2.0581	-0.1708	0.021	-0.002
basic 12 deg 45m/s	3.150	18.462	0.131	0.769
basic 12 deg 90m/s	13.430	75.709	0.1400	0.788
slat 0 deg 45m/s	1.0265	-1.640	0.035	-0.0560
slat 0deg 90 m/s	4.1049	-6.1383	0.035	-0.052
slat 12 deg 45m/s	3.2298	19.959	0.110	0.681
slat 12 deg 90 m/s	13.467	81.642	0.115	0.697

Table 3: Lift and Drag coefficient values based of the 8 different cases

Section 6: Conclusion

Returning to the goal of this report, the slat induced complex effects on the surrounding flow, and thus the rest of the airfoil. Starting with the simplest case for each geometry, (45 m/s and 0 degrees AOA), the slat experienced 5 times more drag than the basic NACA 0015 model. At 0 degrees drag is consistently higher on the slat airfoil model (See table 3). This is likely due to an increase in pressure drag caused by flow separation (see Figure XX).

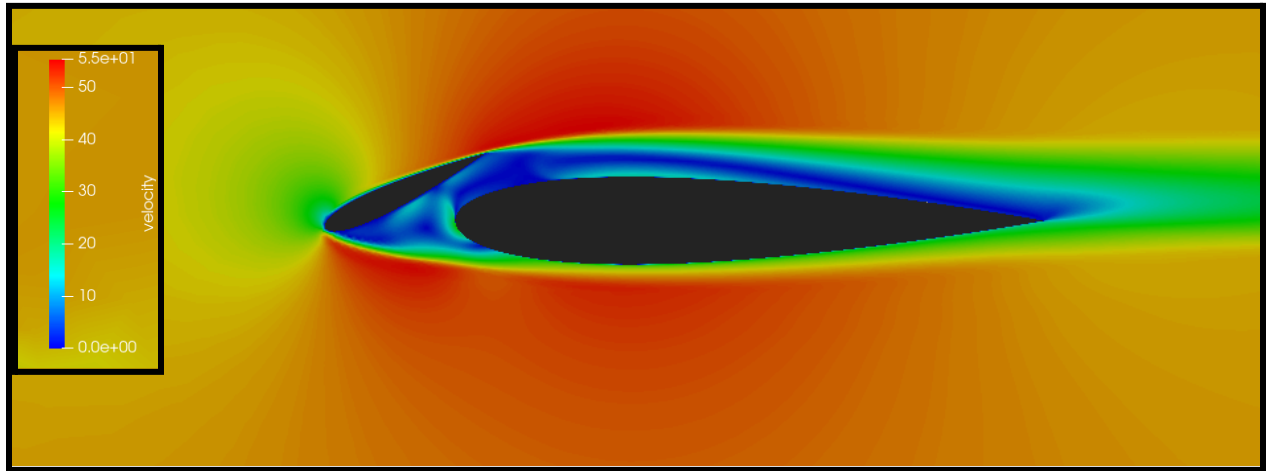


Figure 3: NACA 0015 with LE Slat at 0 AOA 45 m/s

At 12 degrees AOA drag with the slat geometry is only 2% higher than the base case, and C_D follows suit. This is likely due to which geometry is dominant at each angle of attack. At 0 degrees, the slat increases the effective thickness of the wing significantly and is the dominant geometry, thus causing much more drag. At 12 degrees the body of the main wing section is the dominant “thickness” relative to freestream flow direction, reducing the effect the slat has on drag when compared with the basic wing case.

Another interesting note while looking at the 12 degree AOA case is how much less flow separation is induced by the slat (see figure 4 below). As the main wing’s AOA increases, the “relative AOA” of the slat geometry begins to level out, or approach a lower value. Thus, the slat begins to act more like an airfoil and less like a blunt body, leading to more stable flow downstream. This likely reduces the effect of pressure drag when compared with the 0 degree cases.

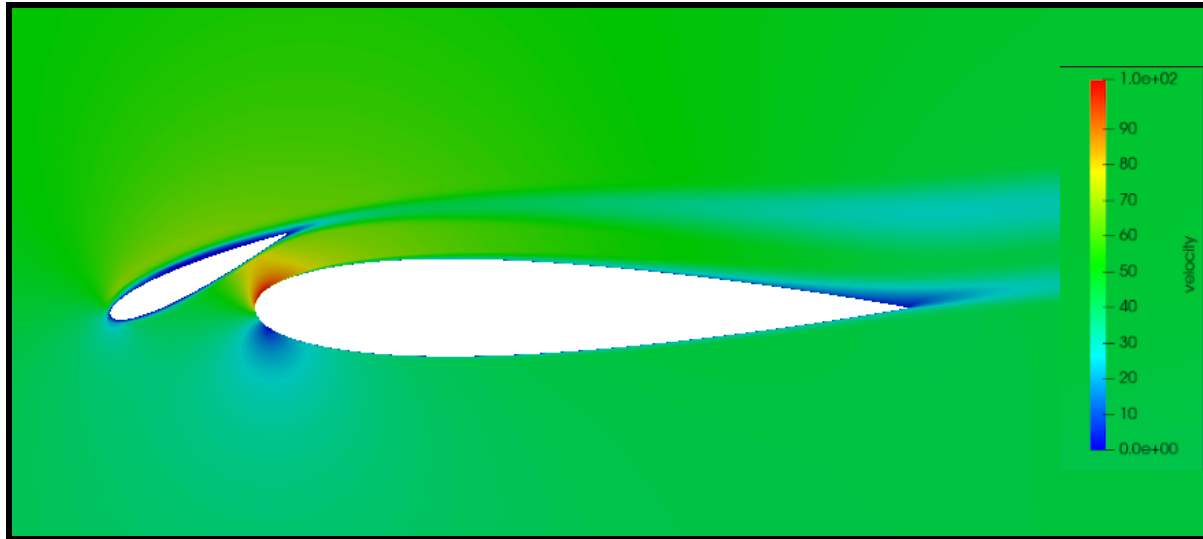


Figure 4: NACA 0015 with LE Slat at 12 AOA 45 m/s

Moving on to lift, the slat geometry actually induced vortices which produced downforce on the wing, reducing lift or even negating it. This phenomenon occurred only at 0 0-degree angle of attack (see figure 3 above). The vortices are present in the slow mixed flow under and past the trailing edge of the slat. Moreover, on the suction side of the main wing the vortices end near the green region along the surface, where flow starts to return to a more recognizable behavior.

At 12 degrees angle of attack, the benefit of the slat becomes evident, with an 8% increase in lift for both speeds when compared with the basic airfoil. Its likely the slat is enabling more favorable flow conditions for higher lift forces as it creates an effective “camber” at higher angles of attack where flow separation is not present (like in Figure 4). The slat keeps the flow from separating.

Overall, at higher angles of attack, slats become the favorable geometry, as they produce significantly more lift and similar drag forces. At 0 AOA, slats perform horribly, inducing flow separation and 5 times more drag. This shows how slats can be used at take off and landing to decrease the stall speed of the aircraft and give more lift at low speeds.

To further explore the efficacy of geometric variation on the NACA 0015 airfoil model, more angles of attack will need to be explored, as well as additional unique geometries.

Appendix:

Basic airfoil velocity photos:

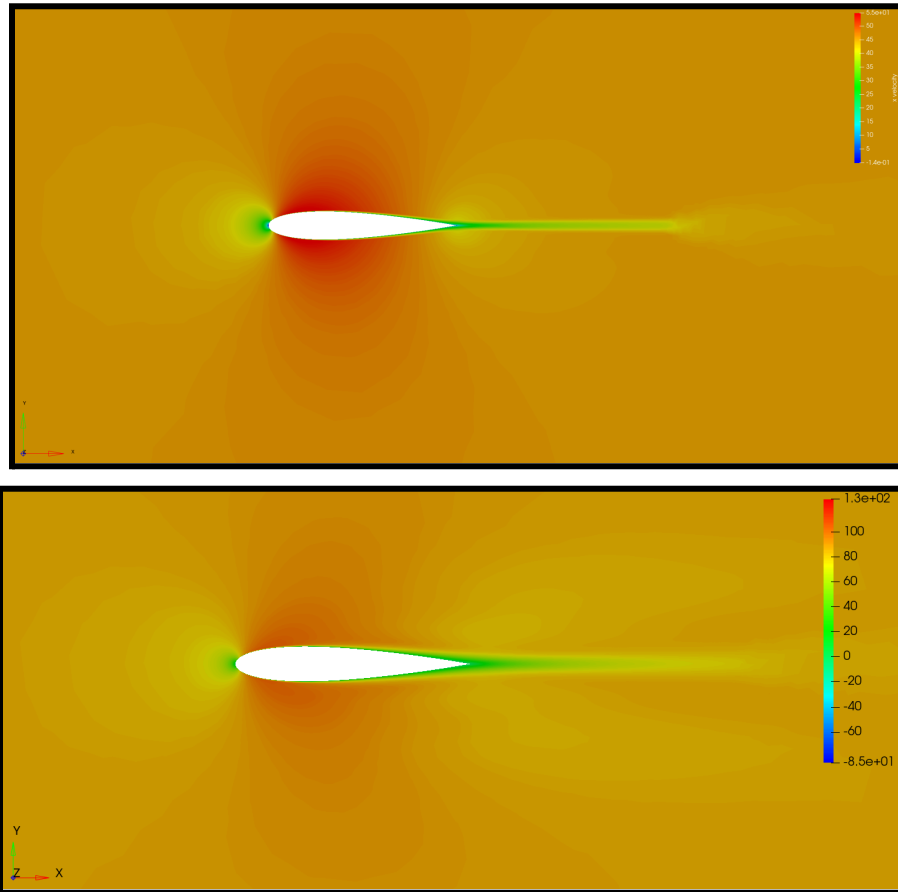


Figure A1: Basic airfoil at 0 deg 45/90 mps

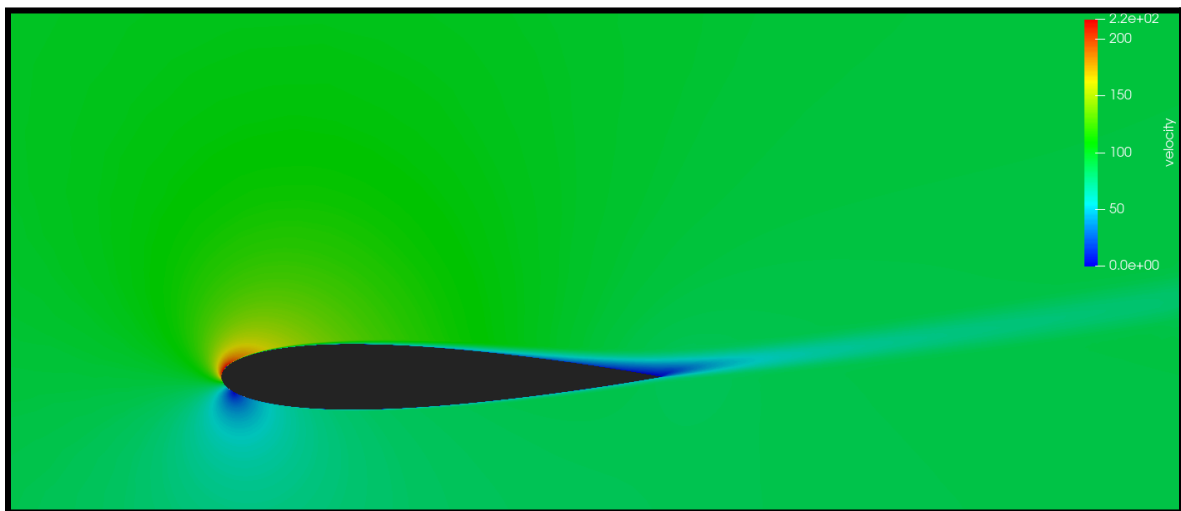


Figure A2: Basic Airfoil at 12 deg 90 mps

Airfoil with slat at 90 m/s:

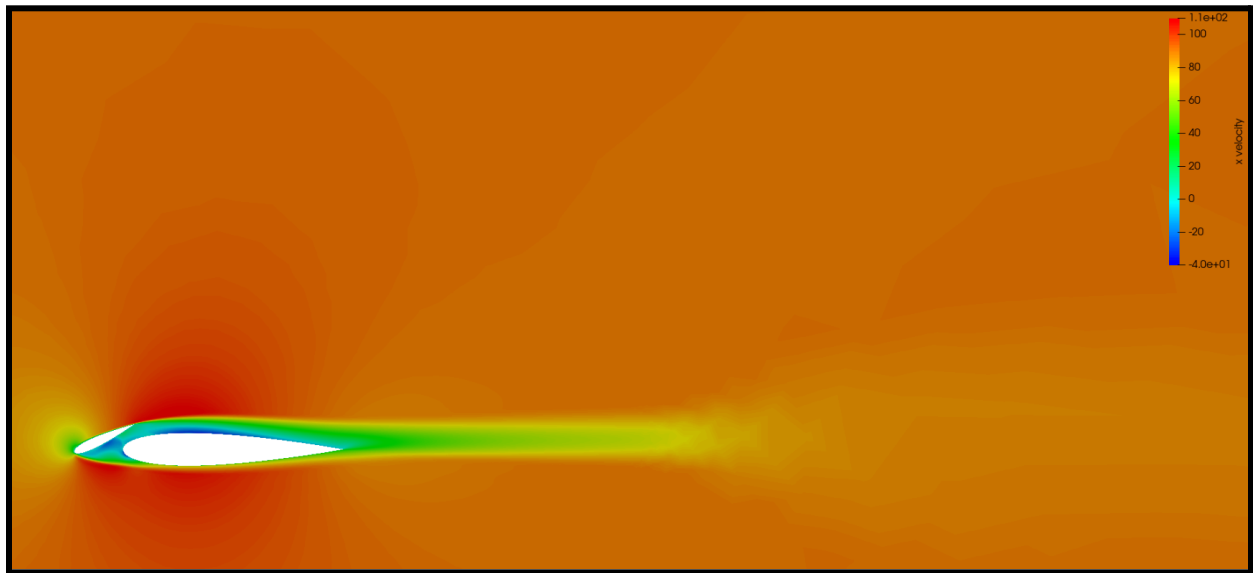


Figure A3: Slat geometry 0 deg 90 m/s

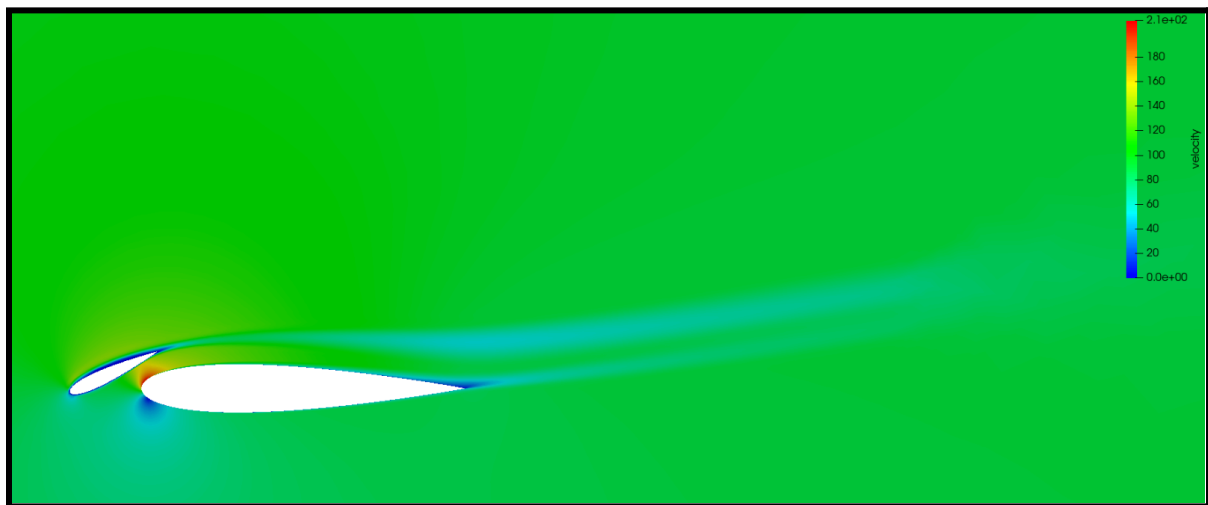


Figure A4: slat geometry 12 deg 90 mps